

DATA

NEXUS

MANIFEST

12 founding principles

BHAGAVAD GITA

Dharma as architecture

8-Layer Protective Architecture Blueprint

Metric	DataNexus	AWS/GCP/Azure	Advantage
Protective layers	8 layers	2–3 layers	DataNexus
Blockchain lineage	Hyperledger Fabric	None	DataNexus
Quality standard	Six Sigma (6σ)	Not defined	DataNexus
Data sovereignty	On-premise, air-gap	Cloud-only	DataNexus
Licensing cost	■0 — open source	■crores/year	DataNexus
NLP queries	Telugu/Hindi/English	English SQL only	DataNexus
Compliance report	60-second proof	Weeks of audit	DataNexus

The Three Unbeatable Pillars

MANIFEST

12 founding principles committed publicly and recorded on blockchain. Data belongs to its creator. Open source forever. Radical transparency always.

BHAGAVAD GITA

Dharma as the operating principle. Nishkama Karma — build for duty, not profit. Viveka — AI that discerns signal from noise. Lokasangraha — for the welfare of all.

SIX SIGMA

3.4 defects per million operations. Every pipeline measured live. 5 sigma minimum SLA guaranteed. Every quality decision logged to Hyperledger Fabric with cryptographic proof.

8

Protective layers

6σ

Quality target

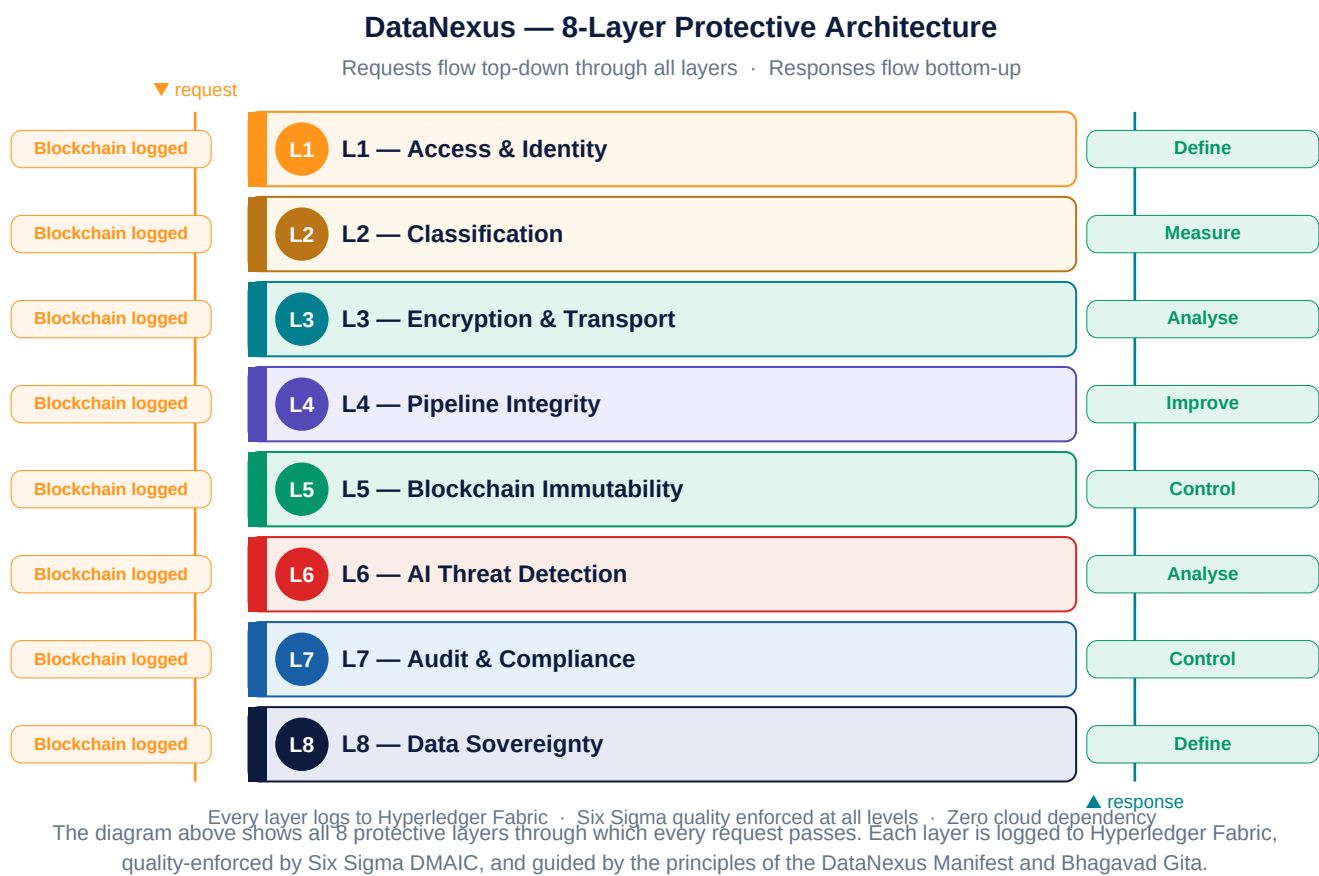
100%

Blockchain logged

0

Cloud dependencies

Architecture Diagram



The 8 Protective Layers — Detail

L1

Access and Identity Layer

MANIFEST — Sovereignty

Tools: [Apache Ranger](#) · [Kerberos](#) · [OAuth2](#) · [MFA](#) · [RBAC](#)

Every request entering DataNexus must pass through identity verification. Kerberos handles cluster-level authentication. OAuth2 handles external API access. Ranger enforces RBAC — who can see which dataset, column, and row. MFA is mandatory for all admin actions. Every login is recorded to Hyperledger Fabric with timestamp, IP, and action. No anonymous access exists anywhere in the system.

Protection: Identity-verified access to every data asset

Six Sigma phase: Define — all access rules formally declared in Ranger before any data flows

Gita: Chapter 6, Verse 5 — the self is both protector and protected

L2

Data Classification and Governance Layer

GITA — Viveka (discernment)

Tools: [Apache Atlas](#) · [Ranger PII masking](#) · [Great Expectations](#) · [DPDP tags](#)

Every dataset is classified automatically on ingestion. Atlas tags data as PII, financial, health, public, or confidential. Ranger applies masking — Aadhaar masked for all except owners, health records column-encrypted. Great Expectations validates schema and content before any data moves. This layer embodies Viveka — wise discernment — applied to data classification.

Protection: Automatic PII masking, DPDP compliance, schema validation

Six Sigma phase: Measure — every dataset scored for quality and classification before processing

Gita: Chapter 2, Verse 50 — skill in action. Discernment is the skill applied to data.

L3

Encryption and Transport Security Layer

SIX SIGMA — Zero Exposure

Tools: [TLS 1.3](#) · [AES-256 at rest](#) · [Kafka SSL](#) · [HDFS encryption zones](#) · [HashiCorp Vault](#)

All data in transit uses TLS 1.3. All data at rest in HDFS uses AES-256 encryption zones, keys managed by HashiCorp Vault. Kafka topics are SSL-encrypted end to end. Key rotation happens automatically every 90 days, logged to Hyperledger Fabric. No plaintext data exists anywhere in the system at any time. Automated scanner verifies every 6 hours.

Protection: AES-256 at rest, TLS 1.3 in transit, auto-rotating keys

Six Sigma phase: Control — encryption verified continuously, every gap logged to Fabric

L4

Pipeline Integrity and Quality Layer

SIX SIGMA — DMAIC Enforced

Tools: [Great Expectations](#) · [Airflow SLA monitor](#) · [Spark validation](#) · [Schema Registry](#)

Every pipeline run is a Six Sigma process. Great Expectations runs quality checks before and after every transformation. Schema Registry enforces contracts — no producer can send a message that breaks a consumer. Airflow SLA monitoring alerts within 60 seconds of any breach. Pipelines falling below 4.5σ are automatically quarantined — data held, not deleted.

Protection: Six Sigma quality gates on every pipeline run

Six Sigma phase: All DMAIC phases — Define contracts, Measure quality, Analyse drops, Improve, Control gates

L5

Blockchain Immutability Layer

ALL THREE — Core Innovation

Tools: [Hyperledger Fabric](#) · [IPFS](#) · [Smart contracts](#) · [SHA-256](#) · [Chaincode](#)

Every transformation, every access, every quality check produces a SHA-256 hash written to Hyperledger Fabric. The hash is permanent — no administrator or employee can alter it. Large datasets are content-addressed on IPFS — the CID is the proof of content. Smart contracts automate compliance: data residency violations are automatically blocked.

Protection: Tamper-proof, permanent audit trail for every data action

Six Sigma phase: Control — blockchain IS the control layer. Every action permanently verifiable.

Gita: Chapter 4, Verse 7 — For the protection of the good, I manifest. Blockchain manifests accountability.

L6

AI Threat Detection and Self-Healing Layer

GITA — Self-Sufficient System

Tools: [Spark MLlib IsolationForest](#) · [Flink CEP](#) · [Self-healing Airflow plugin](#) · [MLflow](#)

An ML model watches every pipeline metric continuously. Flink Complex Event Processing detects attack patterns. Spark MLlib IsolationForest identifies anomalous data access in real time. The self-healing plugin resolves 80%+ of pipeline failures automatically. All AI decisions are logged to Hyperledger Fabric — every automated fix, every threat flag.

Protection: 80%+ self-healing rate, real-time attack detection

Six Sigma phase: Analyse + Improve — AI continuously analyses patterns and improves detection

Gita: Chapter 6, Verse 5 — Let the self lift itself by its own self alone.

L7

Audit, Compliance and Regulatory Layer

MANIFEST — Radical Transparency

Tools: [Apache Atlas lineage](#) · [Ranger audit](#) · [Hyperledger audit API](#) · [DPDP/GDPR reporter](#)

Full audit trail available to any authorised party in real time. A regulator can request a compliance report and receive a cryptographically verified PDF in 60 seconds. DPDP Act and GDPR compliance reports generated automatically on demand. Atlas shows complete data lineage from source through every transform to every consumer.

Protection: 60-second blockchain-verified compliance reports on demand

Six Sigma phase: Control — compliance is a continuous, verifiable state, not a periodic check

Gita: Chapter 10, Verse 41 — Whatever is glorious springs from a spark of truth. Transparency is that truth.

L8

Data Sovereignty and Residency Layer

MANIFEST — Nation First

Tools: [On-premise Hadoop](#) · [Network isolation](#) · [Air-gap mode](#) · [Geo-fencing smart contracts](#)

DataNexus runs entirely on your machines, in your building, in your nation. No data leaves the defined geo-boundary without explicit, blockchain-recorded, multi-signature approval. Air-gap mode allows fully disconnected operation — no internet required, no telemetry, no phone-home. Fabric smart contracts automatically block any pipeline violating geo-residency rules.

Protection: Complete data sovereignty — on-premise, air-gap, geo-fenced

Six Sigma phase: Define — sovereignty is the first definition. Everything is built on top of it.

Gita: Chapter 18, Verse 66 — Surrender to sovereignty, not to vendor dependency.

Open-Source Technology Stack

Every component of DataNexus is Apache-licensed open source — the same foundation AWS, GCP, and Azure built their billion-dollar services on. DataNexus takes the same source and adds blockchain lineage, Six Sigma quality, and AI self-healing — features no cloud vendor has ever offered.

Platform Layer	Open-Source Tool	Replaces (Cloud)	DataNexus Adds
Ingest	Kafka + NiFi + Sqoop	Kinesis/Pub-Sub/Event Hubs	Blockchain event stamp
Store	HDFS + Iceberg + HBase	S3/GCS/ADLS	IPFS content hash
Process	Spark + Flink + YARN	EMR/Dataproc/HDInsight	AI auto-optimizer
Query	Presto + Hive + Druid	Redshift/BigQuery/Synapse	NLP in any language
Orchestrate	Airflow	MWAA/Composer/ADF	Self-healing plugin
Govern	Atlas + Ranger + GE	Lake Formation/Purview	Hyperledger audit
Blockchain	Hyperledger Fabric	None available	Tamper-proof lineage
BI / Dashboards	Superset + Druid	Tableau/Power BI	Sigma score overlay
ML / AI	Spark MLlib + MLflow	SageMaker/Vertex AI	Self-healing models

Six Sigma DMAIC — Applied to Every Layer

Phase	What DataNexus Does	Tools	Blockchain Record
Define	Every pipeline formally declared — owner, Apache Atlas contract	Apache Atlas	Ownership hash on Fabric
Measure	Great Expectations runs quality checks on every Expectations read	Great Expectations + Druid	Quality score on Fabric
Analyse	Spark ML identifies root cause of quality discrepancy	Spark MLlib + MLflow	Diagnosis report on Fabric
Improve	Self-healing plugin applies fix; Spark optimizes slow queries	Airflow + Spark	Fix record on Fabric
Control	Ranger quality gates reject data below 4.5 rating	Ranger + Superset	Every decision on Fabric

9-Month Build Roadmap

Phase 1	Core Hadoop cluster M1–2 · HDFS · YARN · Spark · Kafka · Airflow · NiFi · Presto Complete AWS EMR + Kinesis + Glue + Athena replacement. First DataNexus pipeline running end to end.
Phase 2	Governance and quality M3 · Atlas · Ranger · Great Expectations · Iceberg DPDP Act compliance automatic. Six Sigma dashboard live. PII masking enforced.
Phase 3	Blockchain lineage M4 · Hyperledger Fabric · IPFS · Chaincode · Smart contracts Every transformation hashed. Tamper-proof audit trail live. First blockchain-verified compliance report.
Phase 4	AI and NLP layer M5–6 · LLM NLP · Spark MLlib · MLflow · Druid · Self-healing Airflow NLP queries in Telugu, Hindi, English. 80%+ self-healing pipeline rate. Sub-second OLAP.
Phase 5	UI, pilots and launch M7–9 · React console · Sigma dashboard · Data marketplace · Public beta 3 pilot customers live. GitHub open-source launch. Seed funding closed. Industry shaken.

The Closing Declaration

"DataNexus is the only data platform on earth built on open-source freedom, Dharma as its operating principle, Six Sigma quality with blockchain proof — running on your machines, in your nation, under your law, in your language."

AWS cannot say this. GCP cannot say this. Azure cannot say this. Nobody can say this — except DataNexus. And we can say it because we built the Manifest, we live the Gita, and we enforce Six Sigma with cryptographic proof.

DataNexus · datanexus.io · Hyderabad, India · 2025
Open Source · Dharma-driven · Six Sigma · Blockchain-verified